Light

GCSE Computer Science - Embedded Systems (Set 1)

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GCSE • OCR J277 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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Multiple-choice topic test for **GCSE Computer Science (OCR J277)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

What best describes an embedded system?

- ○ **A.** A computer that has been physically buried underground
- ○ **B.** A computer system designed to perform one or a few dedicated functions, often built into a larger device
- ○ **C.** Any computer connected to the internet
- ○ **D.** A laptop computer used for office work

Q2 (1 mark)

A key characteristic of embedded systems is that they are designed for:

- ○ **A.** Running many different applications chosen by the user
- ○ **B.** Maximum processing power regardless of cost
- ○ **C.** A specific, dedicated purpose rather than general computing tasks
- ○ **D.** Constant internet connectivity

Q3 (1 mark)

A smoke alarm contains a processor that monitors sensors and triggers alerts. This is an example of:

- ○ **A.** A supercomputer
- ○ **B.** An embedded system
- ○ **C.** A gaming console
- ○ **D.** A web server

Q4 (1 mark)

A gaming controller (gamepad) that connects to a console contains processing hardware. What type of system is this?

- ○ **A.** An embedded system
- ○ **B.** A database server
- ○ **C.** A mainframe computer
- ○ **D.** A desktop PC

Q5 (1 mark)

Many embedded systems are designed to use very little electrical power. Why is this important?

- ○ **A.** Low power is never important for embedded systems
- ○ **B.** Many embedded systems run on batteries or must minimise heat generation in enclosed spaces
- ○ **C.** Embedded systems always use more power than desktop computers
- ○ **D.** Power consumption only matters for entertainment devices

Q6 (1 mark)

Most embedded systems have limited or no traditional user interface (keyboard, mouse, monitor). Why?

- ○ **A.** The technology for interfaces doesn't exist
- ○ **B.** All embedded systems must have full keyboards and monitors
- ○ **C.** Users are legally prohibited from interacting with embedded systems
- ○ **D.** They perform their dedicated functions automatically without requiring constant user interaction

Q7 (1 mark)

A central heating boiler contains a controller that manages water temperature, ignition, and safety systems. This controller is:

- ○ **A.** An embedded system

- ○ **B.** A general-purpose computer
- ○ **C.** A smartphone application
- ○ **D.** A cloud server

Q8 (1 mark)

How do embedded systems differ from general-purpose computers like laptops?

- ○ **A.** Embedded systems can run any software the user installs
- ○ **B.** There is no difference between them
- ○ **C.** Embedded systems always have larger screens than laptops
- ○ **D.** Laptops are designed for varied tasks while embedded systems are designed for specific dedicated functions

Q9 (1 mark)

A smart doorbell with a camera, motion detection, and two-way audio contains:

- ○ **A.** An embedded system that processes video, detects motion, and handles communication
- ○ **B.** Only mechanical components with no electronics
- ○ **C.** A full desktop computer
- ○ **D.** No processing capability at all

Q10 (1 mark)

Embedded systems are often designed to operate continuously without crashes or restarts. Why is this important?

- ○ **A.** Reliability doesn't matter for embedded systems
- ○ **B.** Users enjoy restarting their devices frequently
- ○ **C.** Embedded systems are never used in important applications
- ○ **D.** Many embedded systems control critical functions where failures could cause safety issues, financial losses, or major inconvenience

Q11 (1 mark)

A barcode scanner at a shop checkout contains processing hardware that reads codes and communicates with the till system. This is:

- ○ **A.** A video game
- ○ **B.** A word processor
- ○ **C.** An embedded system
- ○ **D.** A spreadsheet application

Q12 (1 mark)

Embedded systems often cost less than general-purpose computers with similar processing power because:

- ○ **A.** They include more features and components
- ○ **B.** General-purpose computers are simpler to design

- ○ **C.** Embedded systems use more expensive materials
- ○ **D.** They only include components necessary for their specific function, reducing manufacturing costs

Q13 (2 marks)

Embedded systems are often physically smaller than general-purpose computers.
Explain why small size is important and how it is achieved.

- ○ **A.** Small size is achieved by removing all electronic components
- ○ **B.** Size has no relevance to embedded system design
- ○ **C.** Small size enables integration into products like wearables and appliances; achieved through integrated microcontrollers and eliminating unnecessary components like expansion slots
- ○ **D.** Small size is impossible for embedded systems

Q14 (2 marks)

A refrigerator's embedded controller monitors temperature sensors and controls the compressor.
Describe two functions this embedded system performs.

- ○ **A.** Playing music and browsing websites
- ○ **B.** Comparing sensor readings to target temperature and cycling the compressor on/off to maintain correct cooling; monitoring door sensors to warn if left open
- ○ **C.** Editing documents and creating presentations
- ○ **D.** Hosting multiplayer games and streaming video

Q15 (2 marks)

Embedded systems often store their programs in firmware rather than loading from a hard drive.
Explain what firmware is and why embedded systems use it.

- ○ **A.** Firmware is physical metal inside the device
- ○ **B.** Firmware is downloaded from the internet each time the device starts
- ○ **C.** Firmware is software permanently stored in ROM or flash memory, providing instant availability at power-on without needing to load from slower storage
- ○ **D.** Firmware is the same as the device's outer casing

Q16 (2 marks)

A speed camera system photographs vehicles exceeding the speed limit. Identify two embedded system functions this requires.

- ○ **A.** Measuring vehicle speed using sensors or radar, and capturing images only when speed exceeds the limit to record the violation
- ○ **B.** Word processing and email management
- ○ **C.** Playing videos and social networking
- ○ **D.** Editing photographs and creating animations

Q17 (2 marks)

Embedded systems often use microcontrollers rather than full computer processors.
Explain what a microcontroller is and why it suits embedded applications.

- ○ **A.** A microcontroller is a tiny human who controls the device
- ○ **B.** A microcontroller is the same as a desktop computer processor
- ○ **C.** A microcontroller is a single chip containing processor, memory, and I/O interfaces, providing a complete system in one component - ideal for embedded systems needing compact, low-cost, low-power solutions
- ○ **D.** A microcontroller can only count to ten

Q18 (2 marks)

A vending machine accepts payment and dispenses products.
Describe two functions its embedded system must perform.

- ○ **A.** Processing payment transactions (coins, notes, cards) and controlling motors to dispense the selected product correctly
- ○ **B.** Running office software and managing spreadsheets
- ○ **C.** Editing videos and creating websites
- ○ **D.** Playing multiplayer games online

Q19 (2 marks)

Embedded systems interact with the physical world through sensors and actuators.
Explain the difference between sensors and actuators.

- ○ **A.** They are different names for the same component
- ○ **B.** Actuators only work during daytime
- ○ **C.** Sensors are always more expensive than actuators

○ **D.** Sensors convert physical conditions (temperature, light, pressure) into electrical signals the system can process; actuators convert electrical signals into physical actions (motors, valves, speakers)

Q20 (2 marks)

A digital alarm clock displays time and sounds an alarm at a set time. Why is an embedded system appropriate for this application?

○ **A.** Clocks require maximum processing power

○ **B.** Clocks must run multiple applications simultaneously

○ **C.** The dedicated functions (timekeeping, display, alarm triggering) need reliable, continuous, low-power operation - characteristics that embedded systems provide efficiently

○ **D.** General-purpose computers are too cheap for clock applications

Q21 (2 marks)

Most embedded systems cannot be easily reprogrammed by end users. Explain why this is typically the case.

○ **A.** Manufacturers forgot to include update features

○ **B.** User modification could cause malfunctions or safety issues, and the dedicated function typically doesn't require software changes

○ **C.** It's illegal for users to understand how embedded systems work

○ **D.** Embedded systems have no memory to store programs

Q22 (2 marks)

Electronic kitchen scales weigh ingredients and display the result. Identify two characteristics that make embedded systems suitable for this application.

- ○ **A.** Need for internet connectivity and social media features
- ○ **B.** Requirement for high-definition video playback
- ○ **C.** Low power consumption (battery operation), instant response to weight changes, and low cost for a consumer product
- ○ **D.** Need to run multiple applications chosen by the user

Q23 (2 marks)

Embedded systems often undergo extensive testing before deployment. Explain why thorough testing is particularly important for embedded systems.

- ○ **A.** Embedded systems often cannot be easily updated after deployment, may control safety-critical functions, and must operate reliably for years
- ○ **B.** Testing is less important for embedded systems than for other software
- ○ **C.** Embedded systems are always perfect without testing
- ○ **D.** Users prefer to find bugs themselves

Q24 (3 marks)

An insulin pump is a wearable medical device that delivers insulin to diabetes patients.
Analyse why embedded system characteristics are critical for this life-sustaining application.

- ○ **A.** The pump needs to play video games between doses
- ○ **B.** The pump requires maximum processing power for graphics
- ○ **C.** Reliability ensures continuous operation (missed doses could be fatal); real-time monitoring adjusts delivery based on sensor readings; small size and low power enable wearable design; security prevents tampering with dosing
- ○ **D.** Users need to install their own medical software

Q25 (3 marks)

An industrial robot arm in a factory performs repetitive assembly tasks. Analyse the embedded system requirements for this application.

- ○ **A.** The robot only needs to display advertisements
- ○ **B.** Industrial robots run standard office software
- ○ **C.** Robots don't use any electronic control systems
- ○ **D.** Precise motion control requires real-time processing; safety systems must instantly stop movement when sensors detect humans; reliability ensures continuous production; programmable paths allow flexibility without full general-purpose computing

Q26 (3 marks)

A company must decide whether to use an embedded system or a general-purpose computer for a new product.
Analyse the key factors that would favour choosing an embedded system.

- ○ **A.** Dedicated function that won't change, need for low cost per unit at high volumes, requirements for low power consumption, small physical size, high reliability, and instant boot time

○ **B.** Need to run many different applications, maximum flexibility, and frequent software changes by users

○ **C.** Requirement for large monitor displays and full keyboards

○ **D.** Need for users to install their own software and games

Q27 (3 marks)

A smart door lock allows entry using PIN codes, fingerprints, and smartphone apps.
Analyse the embedded system requirements and security considerations for this application.

○ **A.** The lock only needs to look attractive without any electronic functions

○ **B.** The embedded system must process biometric data, communicate securely with smartphones, store authorised credentials safely, and resist physical and electronic tampering - a compromise could allow burglary

○ **C.** Smart locks don't require any processing capability

○ **D.** Security is unimportant for door locks

Q28 (3 marks)

A wireless environmental sensor monitors temperature and humidity, reporting via radio to a central system.
Analyse how power management enables years of battery operation.

○ **A.** The embedded system uses sleep modes between readings, wakes briefly to measure and transmit, uses low-power components, and minimises radio transmission time - enabling years of operation on small batteries

○ **B.** The sensor uses maximum power continuously to ensure fast response

○ **C.** Battery life doesn't matter for sensors

○ **D.** Sensors must be plugged into mains power at all times

Q29 (3 marks)

A television set-top box receives digital broadcasts, decodes video, and outputs to the TV.
Analyse the embedded system requirements for this application.

○ **A.** Set-top boxes only need to display static images

○ **B.** Real-time video decoding requires significant processing power; the system must handle multiple broadcast standards; interactive features need user interface processing; continuous operation demands reliability - all achieved with dedicated embedded design

○ **C.** Set-top boxes are simple devices requiring no processing

○ **D.** They only operate with physical video tapes

Q30 (3 marks)

The Internet of Things (IoT) connects embedded systems to networks and cloud services.
Analyse the opportunities and challenges this creates for embedded systems.

○ **A.** IoT has no effect on embedded systems

○ **B.** IoT only applies to desktop computers

○ **C.** Connecting embedded systems to the internet makes them completely secure

○ **D.** Opportunities: remote monitoring, data analytics, automatic updates, and enhanced functionality. Challenges: security vulnerabilities from network exposure, privacy concerns from data collection, power consumption from connectivity, and complexity from cloud integration

Total: **55** marks • Time: **30 min**

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